

Learning Dexterous In-Hand Reorientation of a Numbered Dodecahedron

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Why

Dexterous in-hand manipulation allows a robotic hand to change an object’s orientation without placing it down or using external support. This capability is useful in assembly, tool use, inspection, and object handover. It is also challenging: many finger joints must be coordinated while fingertip contacts appear, disappear, and move. A small change in finger motion may cause slipping, an unintended collision, or an object drop.

A regular dodecahedron provides an interesting setting for this problem. Its twelve faces create many distinct target orientations, while its edges and vertices produce frequent contact transitions during rotation. We label the faces from 1 to 12 and assign every label a fixed upright direction. The labels define a unique object coordinate frame; they are not visual inputs to the policy, and the task is not limited to twelve “face-up” poses. Instead, the object may be commanded to any orientation in $SO(3)$.

Model-based planning and control provide an explicit way to incorporate object geometry, dynamics, and contact constraints, and can generate interpretable motions when the model and contact sequence are known. However, planning through changing contacts remains difficult and is sensitive to errors in friction, contact parameters, and object pose. Reinforcement learning (RL) offers a complementary approach: an agent can discover low-level finger coordination through repeated interaction and learn a feedback policy that reacts to the observed hand and object states. We will investigate whether a dexterous hand can learn arbitrary reorientation of a numbered dodecahedron and whether gradually increasing target difficulty improves learning efficiency and manipulation reliability.

Problem Statement

We consider a fixed-base Wuji right hand with 20 independently controlled joints; wrist or robot-arm motion is outside the present task. At reset, the hand and dodecahedron are sampled jointly from feasible initial grasp configurations, rather than arbitrarily rotating the object inside a fixed hand pose. Object size, mass, and contact properties remain fixed in the main experiments so that the learning strategies can be compared under the same physical conditions.

The agent receives a desired orientation $R_g \in SO(3)$ relative to the palm and must rotate the object toward it without losing the grasp. We measure the orientation error using the geodesic distance

$$e_R = \cos^{-1} \left(\frac{\text{tr}(R_g^\top R_o) - 1}{2} \right), \quad (1)$$

where R_o is the current object orientation. A target is reached when $e_R < 0.2$ rad for five consecutive control steps. After a successful hold, a new target is generated, allowing multiple rotations within an episode. An episode terminates when the object remains outside the hand’s retention region or the time limit is reached.

Initial development and evaluation will be conducted in simulation for controlled, repeatable experiments. The actor will use joint and object-pose observations that can in principle be measured on a physical system, preserving physical deployment as a possible later extension.

RL Cast

Policy observations. The actor receives a three-step history containing:

- the 20 normalized joint positions and their errors from the commanded joint positions;
- the object position and current orientation relative to the palm, with orientation represented continuously in 6D;
- the 6D relative rotation from the current to the target orientation; and
- the previous 20-dimensional action.

Including both current object orientation and goal error lets the policy distinguish, for example, a face contact from an edge contact even when both states require the same target rotation. During training, the critic may also use

privileged simulation quantities such as joint velocities, fingertip positions, object velocities, applied disturbances, and physics parameters. These quantities are not given to the actor.

Action space. The action $a_t \in [-1, 1]^{20}$ specifies offsets to the 20 joint-position targets. Offsets are scaled and passed through an exponential moving-average filter before reaching the simulated actuators. The policy acts at 20 Hz. This low-level action space leaves finger coordination, contact redistribution, and grasp retention to the learned policy.

Reward structure. With all weights positive, the per-step reward is

$$r_t = w_o r_{\text{ori}} + w_h r_{\text{hold}} + w_d r_{\text{detach}} - w_p c_{\text{pose}} - w_a c_{\text{action}} - w_\tau c_{\text{torque}} - w_s c_{\text{slide}} - w_e c_{\text{escape}} - w_c c_{\text{collision}}. \quad (2)$$

r_{ori} is one inside the angular tolerance and otherwise decreases linearly with e_R , providing dense progress feedback. After five consecutive steps inside the tolerance, r_{hold} increases while the object remains accurate. The optional r_{detach} rewards support by distal finger links without palm or proximal-link contact. The costs penalize deviation from a nominal hand pose, rapid action changes, actuator effort, relative fingertip sliding during contact, escape from the retention region, and finger self-collision. Sustained escape also terminates the episode. Reward weights will be initialized from an existing implementation and treated as hyperparameters. Overall, the reward encourages accurate reorientation while discouraging object drops and unnecessarily aggressive motions.

RL Algorithm and Experiments

We will train an asymmetric actor-critic policy using Proximal Policy Optimization (PPO). The actor and critic are multilayer perceptrons with ELU activations and observation normalization. Thousands of MuJoCo environments will run in parallel on a GPU to collect contact-rich experience.

We compare two strategies under identical environment-step budgets and physical parameters:

- **Uniform-goal training:** targets cover the complete $SO(3)$ space from the start.
- **Goal-difficulty curriculum:** targets initially differ from the current orientation by at most 30° . This limit increases with recent success until targets cover the complete $SO(3)$ space.

Each strategy will use at least three random seeds. For the primary evaluation, each trial resets to one held-out feasible grasp and one held-out uniformly sampled target. A trial succeeds only if the target is reached within a common time limit; drops and timeouts count as failures. Target success rate is $N_{\text{success}}/N_{\text{trials}}$. We will also report drop rate, minimum orientation error, time to success, performance versus target angle, and learning curves versus environment steps. A separate continuous-goal test will report goals completed per episode and the fraction of episodes ending in a drop.

We expect the curriculum may reduce early drops by first teaching stable local contact adjustments, but it may also over-specialize to small rotations and offer little final benefit. The experiments will determine which effect dominates and identify failure modes involving particular faces, edges, and large-angle rotations.